

Elementare Gleichungen - Aufgaben

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1 Einführung

1.1 Aufgaben

I. Löse die folgenden Gleichungen:

1. Löse die Gleichung $2x + 4 = 18$.

$$\begin{array}{rcl} 2x + 4 = 18 & | & - 4 \\ 2x = 14 & | & \div 2 \\ x = \mathbf{7} \end{array}$$

2. Löse die Gleichung $6x + 3 = 33$.

$$\begin{array}{rcl} 6x + 3 = 33 & | & - 3 \\ 6x = 30 & | & \div 6 \\ x = \mathbf{5} \end{array}$$

3. Löse die Gleichung $5x + 8 = 13$.

$$\begin{array}{rcl} 5x + 8 = 13 & | & - 8 \\ 5x = 5 & | & \div 5 \\ x = \mathbf{1} \end{array}$$

4. Löse die Gleichung $3x + 9 = 18$.

$$\begin{array}{rcl} 3x + 9 = 18 & | & - 9 \\ 3x = 9 & | & \div 3 \\ x = \mathbf{3} \end{array}$$

5. Löse die Gleichung $6x + 9 = 45$.

$$\begin{array}{rcl} 6x + 9 = 45 & | & - 9 \\ 6x = 36 & | & \div 6 \\ x = \mathbf{6} \end{array}$$

6. Löse die Gleichung $4x + 9 = 29$.

$$\begin{array}{rcl} 4x + 9 = 29 & | & - 9 \\ 4x = 20 & | & \div 4 \\ x = \mathbf{5} \end{array}$$

7. Löse die Gleichung $7x + 4 = 25$.

$$\begin{array}{rcl} 7x + 4 = 25 & | & - 4 \\ 7x = 21 & | & \div 7 \\ x = \mathbf{3} \end{array}$$

8. Löse die Gleichung $2x + 7 = 11$.

$$\begin{array}{rcl} 2x + 7 = 11 & | & - 7 \\ 2x = 4 & | & \div 2 \\ x = \mathbf{2} \end{array}$$

9. Löse die Gleichung $4x + 8 = 36$.

$$\begin{array}{rcl} 4x + 8 = 36 & | & - 8 \\ 4x = 28 & | & \div 4 \\ x = \mathbf{7} \end{array}$$

10. Löse die Gleichung $6x + 4 = 46$.

$$\begin{array}{rcl} 6x + 4 = 46 & | & - 4 \\ 6x = 42 & | & \div 6 \\ x = \mathbf{7} \end{array}$$

2 Negative Zahlen

2.1 Aufgaben zu Kapitel 2.1 bis 2.3

- Was sind negative Zahlen?
- Negative Zahlen im Alltag
- Addition mit negativen Zahlen

I. Berechne die folgenden Summen:

1. $4 + (-3) = 4 - 3 = \mathbf{1}$
2. $5 + (-2) = 5 - 2 = \mathbf{3}$
3. $4 + (-6) = 4 - 6 = \mathbf{-2}$
4. $9 + (-7) = 9 - 7 = \mathbf{5}$
5. $8 + (-3) = 8 - 3 = \mathbf{5}$
6. $4 + (-17) = 4 - 17 = \mathbf{-13}$
7. $14 + (-5) = 14 - 5 = \mathbf{9}$
8. $1 + (-8) = 1 - 8 = \mathbf{-7}$
9. $9 + (-12) = 9 - 12 = \mathbf{-3}$
10. $7 + (-20) = 7 - 20 = \mathbf{-13}$
11. $23 + (-12) = 23 - 12 = \mathbf{11}$
12. $33 + (-21) = 33 - 21 = \mathbf{12}$
13. $14 + (-6) = 14 - 6 = \mathbf{8}$
14. $25 + (-30) = 25 - 30 = \mathbf{-5}$

15. $5 + (-9) = 5 - 9 = -4$
16. $17 + (-8) = 17 - 8 = 9$
17. $11 + (-18) = 11 - 18 = -7$
18. $17 + (-25) = 17 - 25 = -8$
19. $27 + (-6) = 27 - 6 = 21$
20. $13 + (-43) = 13 - 43 = -30$
21. $4 + (-1) = 4 - 1 = 3$
22. $-2 + 3 = 1$
23. $-2 + (-4) = -2 - 4 = -6$
24. $3 + (-5) = 3 - 5 = -2$
25. $-2 + 1 = -1$
26. $-5 + (-3) = -5 - 3 = -8$
27. $4 + (-5) = 4 - 5 = -1$
28. $-3 + (-4) = -3 - 4 = -7$
29. $-2 + (-7) = -2 - 7 = -9$
30. $2 + (-8) = 2 - 8 = -6$
31. $-1 + (-1) = -1 - 1 = -2$
32. $4 + (-7) = 4 - 7 = -3$
33. $-2 + (-3) = -2 - 3 = -5$
34. $-3 + 5 = 2$
35. $-3 + (-4) = -3 - 4 = -7$
36. $-1 + (-7) = -1 - 7 = -8$
37. $-3 + 4 = 1$
38. $2 + (-6) = 2 - 6 = -4$
39. $-3 + 2 = -1$
40. $-2 + (-8) = -2 - 8 = -10$
41. $2 + (-2) = 2 - 2 = 0$
42. $-1 + 1 = 0$
43. $-3 + 3 = 0$
44. $5 + (-5) = 5 - 5 = 0$

45. $-7 + 7 = 0$

46. $13 + (-13) = 13 - 13 = 0$

47. $-8 + 8 = 0$

48. $35 + (-35) = 35 - 35 = 0$

II. Addiere die benachbarten Zahlen und schreibe das Ergebnis darüber. Mithilfe der obersten Zahl kann man erkennen, ob man richtig gerechnet hat.

a)

			-6		
		-4		-2	
	-1		-3		1
1		-2		-1	2
-2	3		-5	4	-2

b)

			-8		
		-1		-7	
	1		-2		-5
1,5		-0,5		-1,5	-3,5
3,5	-2		1,5	-3	-0,5

2.2 Aufgaben zu Kapitel 2.4 - Subtraktion mit negativen Zahlen

I. Berechne die folgenden Differenzen:

1. $4 - 3 = 1$

2. $8 - 5 = 3$

3. $5 - 7 = -2$

4. $10 - 7 = 3$

5. $6 - 19 = -13$

6. $13 - 5 = 8$

7. $2 - 9 = -7$

8. $11 - 14 = -3$

9. $6 - 30 = -24$

10. $34 - 23 = 11$

11. $15 - 7 = 8$

12. $36 - 41 = -5$
13. $4 - 8 = -4$
14. $12 - 17 = -5$
15. $18 - 26 = -8$
16. $26 - 5 = 21$
17. $15 - 45 = -30$
18. $-9 - (-2) = -9 + 2 = -7$
19. $3 - (-2) = 3 + 2 = 5$
20. $-4 - (-1) = -4 + 1 = -3$
21. $-2 - 3 = -5$
22. $-2 - 4 = -6$
23. $3 - (-5) = 3 + 5 = 8$
24. $-2 - 1 = -3$
25. $-5 - 3 = -8$
26. $-3 - (-4) = -3 + 4 = 1$
27. $-2 - (-4) = -2 + 4 = 2$
28. $-1 - (-1) = -1 + 1 = 0$
29. $4 - (-7) = 4 + 7 = 11$
30. $-2 - (-3) = -2 + 3 = 1$
31. $-3 - 5 = -8$
32. $-3 - (-4) = -3 + 4 = 1$
33. $-1 - 7 = -8$
34. $-3 - 4 = -7$
35. $2 - (-6) = 2 + 6 = 8$
36. $-3 - 2 = -5$
37. $-2 - (-8) = -2 + 8 = 6$
38. $-2 - 2 = -4$
39. $-1 - 1 = -2$
40. $4 - (-4) = 4 + 4 = 8$
41. $7 - (-7) = 7 + 7 = 14$

$$42. \quad 35 - (-35) = 35 + 35 = \mathbf{70}$$

$$43. \quad 3 - 5 = \mathbf{-2}$$

$$44. \quad -4 - 6 = \mathbf{-10}$$

$$45. \quad -9 - (-6) = -9 + 6 = \mathbf{-3}$$

$$46. \quad -4 - (-8) = -4 + 8 = \mathbf{4}$$

$$47. \quad 5 - (-4) = 5 + 4 = \mathbf{9}$$

$$48. \quad 4 - 7 = \mathbf{-3}$$

II. Subtrahiere die benachbarten Zahlen (linke Zahl minus rechte Zahl) und schreibe das Ergebnis darüber. Mithilfe der obersten Zahl kann man erkennen, ob man richtig gerechnet hat.

$$\begin{array}{cccccccc} & & & & -62 & & & \\ & & & & -30 & 32 & & \\ a) & & & -13 & 17 & -15 & & \\ & -5 & & 8 & -9 & 6 & & \\ & -2 & 3 & & -5 & 4 & -2 & \end{array}$$

$$\begin{array}{cccccccc} & & & & 32 & & & \\ & & & 17 & -15 & & & \\ b) & & 9 & -8 & 7 & & & \\ & 5,5 & -3,5 & 4,5 & -2,5 & & & \\ & 3,5 & -2 & 1,5 & -3 & -0,5 & & \end{array}$$

2.3 Aufgaben zu Kapitel 2.5

- Verschmelzung von Addition und Subtraktion

I. Berechne die folgenden Summen/Differenzen:

$$1. \quad -27 + 8 = \mathbf{-19}$$

$$2. \quad -18 - 13 = \mathbf{-31}$$

$$3. \quad -28 - 1 = \mathbf{-29}$$

$$4. \quad -24 + 17 = \mathbf{-7}$$

$$5. \quad -7 - 12 = \mathbf{-19}$$

$$6. \quad -11 + 17 = \mathbf{6}$$

$$7. \quad -21 + 15 = \mathbf{-6}$$

8. $17 - 15 = \mathbf{2}$
9. $-16 - 17 = \mathbf{-33}$
10. $-29 + 7 = \mathbf{-22}$
11. $-11 - 23 = \mathbf{-34}$
12. $6 - 30 = \mathbf{-24}$
13. $-5 + 2 = \mathbf{-3}$
14. $2 - 11 = \mathbf{-9}$
15. $11 - 26 = \mathbf{-15}$
16. $-16 - 16 = \mathbf{-32}$
17. $6 - 23 = \mathbf{-17}$
18. $21 - 22 = \mathbf{-1}$
19. $-28 - 13 = \mathbf{-41}$
20. $29 - 22 = \mathbf{7}$
21. $5 - 16 = \mathbf{-11}$
22. $-24 - 11 = \mathbf{-35}$
23. $-7 - 5 = \mathbf{-12}$
24. $-13 - 26 = \mathbf{-39}$
25. $-1 - 12 = \mathbf{-13}$
26. $-3 - 5 = \mathbf{-8}$
27. $-21 + 8 = \mathbf{-13}$
28. $-16 + 12 = \mathbf{-4}$
29. $-6 - 1 = \mathbf{-7}$
30. $-3 + 7 = \mathbf{4}$

2.4 Aufgaben zu Kapitel 2.6 bis 2.8

- Addition und Subtraktion mehrerer Zahlen
- Klammern subtrahieren und addieren
- Beispiele zur Addition und Subtraktion

I. Berechne die folgenden Summen/Differenzen:

1. $-1 + 5 - 3 = 4 - 3 = \mathbf{1}$
2. $1 - 5 - 3 = -4 - 3 = \mathbf{-7}$
3. $1 + 5 - 3 = 6 - 3 = \mathbf{3}$

4. $-1 - 5 - 3 = -6 - 3 = \mathbf{-9}$
5. $-1 + 5 + 3 = 4 + 3 = \mathbf{7}$
6. $1 - 5 + 3 = -4 + 3 = \mathbf{-1}$
7. $1 + 5 + 3 = 6 + 3 = \mathbf{9}$
8. $-1 - 5 + 3 = -6 + 3 = \mathbf{-3}$
9. $-2 + 7 + 4 = 5 + 4 = \mathbf{9}$
10. $-4 + 2 - 7 = -2 - 7 = \mathbf{-9}$
11. $-7 + 4 - 2 = -3 - 2 = \mathbf{-5}$
12. $-5 - 6 - 2 = -11 - 2 = \mathbf{-13}$
13. $2 - 6 - 3 = -4 - 3 = \mathbf{-7}$
14. $-4 - 1 + 4 = -5 + 4 = \mathbf{-1}$
15. $7 + 3 - 2 = 10 - 2 = \mathbf{8}$
16. $-2 + 7 + 1 = 5 + 1 = \mathbf{6}$
17. $1 - 4 + 5 = -3 + 5 = \mathbf{2}$
18. $-7 + 2 - 9 = -5 - 9 = \mathbf{-14}$
19. $-2 - 5 - 6 = -7 - 6 = \mathbf{-13}$
20. $1 - 5 - 8 = -4 - 8 = \mathbf{-12}$
21. $8 - 5 + 7 - 20 = 3 + 7 - 20 = 10 - 20 = \mathbf{-10}$
22. $-9 - 4 + 11 - 17 = -13 + 11 - 17 = -2 - 17 = \mathbf{-19}$

II. Berechne die folgenden Summen/Differenzen:

1. $-16 - 6 + 19 = -22 + 19 = \mathbf{-3}$
2. $-5 + 15 + 2 - 13 + 9 = 10 + 2 - 13 + 9 = 12 - 13 + 9 = -1 + 9 = \mathbf{8}$
3. $-4 - 10 - 18 - 13 + 8 = -14 - 18 - 13 + 8 = -32 - 13 + 8 = -45 + 8 = \mathbf{-37}$
4. $-18 - 6 - 7 - 9 + 15 = -24 - 7 - 9 + 15 = -31 - 9 + 15 = -40 + 15 = \mathbf{-25}$
5. $-17 + 14 - 9 = -3 - 9 = \mathbf{-12}$
6. $-20 - 1 + 3 = -21 + 3 = \mathbf{-18}$
7. $16 - 12 + 13 - 29 = 4 + 13 - 29 = 17 - 29 = \mathbf{-12}$
8. $-2 - 29 + 6 = -31 + 6 = \mathbf{-25}$
9. $-17 + 14 - 9 = -3 - 9 = \mathbf{-12}$
10. $-10 - 3 - 27 = -13 - 27 = \mathbf{-40}$

11. $-19 - 13 + 5 - 1 - 24 = -32 + 5 - 1 - 24 = -27 - 1 - 24 = -28 - 24 = -\mathbf{52}$
12. $-21 - 2 + 11 = -23 + 11 = -\mathbf{12}$
13. $-25 - 4 + 13 = -29 + 13 = -\mathbf{16}$
14. $-1 - 17 + 20 - 14 = -18 + 20 - 14 = 2 - 14 = -\mathbf{12}$
15. $12 + 27 - 3 = 39 - 3 = \mathbf{36}$
16. $23 - 26 - 14 + 8 + 27 = -3 - 14 + 8 + 27 = -17 + 8 + 27 = -9 + 27 = \mathbf{18}$
17. $26 - 24 + 16 - 5 - 10 = 2 + 16 - 5 - 10 = 18 - 5 - 10 = 13 - 10 = \mathbf{3}$
18. $-6 + 13 - 11 = 7 - 11 = -\mathbf{4}$
19. $17 - 7 + 8 = 10 + 8 = \mathbf{18}$
20. $-21 + 3 + 5 + 16 + 10 = -18 + 5 + 16 + 10 = -13 + 16 + 10 = 3 + 10 = \mathbf{13}$
21. $14 - 1 + 10 - 22 - 1 = 13 + 10 - 22 - 1 = 23 - 22 - 1 = 1 - 1 = \mathbf{0}$
22. $-6 - 11 + 17 - 12 + 13 = -17 + 17 - 12 + 13 = -12 + 13 = \mathbf{1}$
23. $6 + 29 - 13 = 35 - 13 = \mathbf{22}$
24. $-16 - 23 - 25 = -39 - 25 = -\mathbf{64}$
25. $-18 + 9 - 20 = -9 - 20 = -\mathbf{29}$
26. $-13 + 22 + 7 + 21 = 9 + 7 + 21 = 16 + 21 = \mathbf{37}$
27. $1 + 10 - 22 + 6 - 11 = 11 - 22 + 6 - 11 = -11 + 6 - 11 = -5 - 11 = -\mathbf{16}$
28. $16 - 17 + 25 = -1 + 25 = \mathbf{24}$
29. $-5 - 29 - 8 - 23 = -34 - 8 - 23 = -42 - 23 = -\mathbf{65}$
30. $-13 + 4 - 12 - 7 = -9 - 12 - 7 = -21 - 7 = -\mathbf{28}$

III. Berechne:

1. $8 - (9 - 4) = 8 - 5 = \mathbf{3}$
2. $13 - (3 - 7) = 13 - (-4) = 13 + 4 = \mathbf{17}$
3. $12 - (-2 - 6) = 12 - (-8) = 12 + 8 = \mathbf{20}$
4. $7 - (-4 + 6) = 7 - 2 = \mathbf{5}$
5. $1 + (3 - 7) = 1 + (-4) = 1 - 4 = -\mathbf{3}$
6. $4 + (-3 - 7) = 4 + (-10) = 4 - 10 = -\mathbf{6}$
7. $0 - (8 - 3) = 0 - 5 = -\mathbf{5}$
8. $0 - (-4 - 9) = 0 - (-13) = \mathbf{13}$
9. $-(-2 + 6) = -\mathbf{4}$

10. $-(4 + 6) = -\mathbf{10}$
11. $-(5 - 12) = -(-7) = \mathbf{7}$
12. $-(-3 - 7) - (8 - 5) = -(-10) - 3 = 10 - 3 = \mathbf{7}$
13. $6 - (8 - 5) - 3 = 6 - 3 - 3 = 3 - 3 = \mathbf{0}$
14. $-(4 - 7) - (-3 - 2) = -(-3) - (-5) = 3 + 5 = \mathbf{8}$
15. $-(-3 + 7) + (8 - 5) + 3 = -4 + 3 + 3 = -1 + 3 = \mathbf{2}$
16. $(4 - 8) - (9 - 6) = -4 - 3 = -\mathbf{7}$
17. $-(-3 + 11) + (8 - 5) = -8 + 3 = -\mathbf{5}$
18. $-(-2 - 6) - (13 - 7) = -(-8) - 6 = 8 - 6 = \mathbf{2}$
19. $-(9 + 4) - (-11 + 17) = -13 - 6 = -\mathbf{19}$
20. $-(9 - 4 + 11 - 17) = -(5 + 11 - 17) = -(16 - 17) = -(-1) = \mathbf{1}$
21. $7 - (4 + 6) = 7 - 10 = -\mathbf{3}$
22. $-(2 - (6 - 8)) = -(2 - (-2)) = -(2 + 2) = -\mathbf{4}$

IV. Schreibe übersichtlich und berechne:

1. $3 - (7 - (10 - 4)) = 3 - (7 - 6) = 3 - 1 = \mathbf{2}$
2. $14 - (5 - (8 - 3)) = 14 - (5 - 5) = 14 - 0 = \mathbf{14}$
3. $12 - (-4 - (-2 - 6)) = 12 - (-4 - (-8)) = 12 - (-4 + 8) = 12 - 4 = \mathbf{8}$
4. $3 - (-(-4 + 6) + 8) = 3 - (-2 + 8) = 3 - 6 = -\mathbf{3}$
5. $9 - (- (3 - 7) - (-5 + 13)) = 9 - (-(-4) - 8) = 9 - (4 - 8) = 9 - (-4) = 9 + 4 = \mathbf{13}$
6. $-(4 - (6 - 7)) + 7 = -(4 - (-1)) + 7 = -(4 + 1) + 7 = -5 + 7 = \mathbf{2}$
7. $(-4 - 9) - (8 - 3) = -13 - 5 = -\mathbf{18}$
8. $-(-5 + 3) + (7 - 18) = -(-2) + (-11) = 2 - 11 = -\mathbf{9}$
9. $-((-5 + 3) + (7 - 18)) = -(-2 + (-11)) = -(-2 - 11) = -(-13) = \mathbf{13}$
10. $3 - (4 - (5 - (-7 + 8))) = 3 - (4 - (5 - 1)) = 3 - (4 - 4) = 3 - 0 = \mathbf{3}$
11. $(-8 + 15) - (5 - (6 - 18)) = 7 - (5 - (-12)) = 7 - (5 + 12) = 7 - 17 = -\mathbf{10}$
12. $-((-4 - 9) - (12 - 7)) = -(-13 - 5) = -(-18) = \mathbf{18}$
13. $6 - ((8 - 5) - 3) = 6 - (3 - 3) = 6 - 0 = \mathbf{6}$
14. $1 + ((- (4 - 7) + 3) - (-1 - 2)) = 1 + ((-(-3) + 3) - (-1 - 2)) = 1 + ((3 + 3) - (-3)) = 1 + (6 + 3) = 1 + 9 = \mathbf{10}$
15. $(-(-5)) + (-5) = 5 - 5 = \mathbf{0}$

$$16. \quad 4 - \left(- \left(- (-1) \right) \right) = 4 - (-1) = 4 + 1 = \mathbf{5}$$

$$17. \quad 5 + (4 - (-3 + 11)) = 5 + (4 - 8) = 5 + (-4) = 5 - 4 = \mathbf{1}$$

$$18. \quad 4 - (-3 - 9) - (12 - 25) = 4 - (-12) - (-13) = 4 + 12 + 13 = 16 + 13 = \mathbf{29}$$

$$19. \quad 5 - \left((-3 + 2) - (7 + (-9)) \right) = 5 - (-1 - (7 - 9)) = 5 - (-1 - (-2)) = 5 - (-1 + 2) = 5 - 1 = \mathbf{4}$$

$$20. \quad 1 - \left(2 - 3 + \left(4 - (5 - 6 + (7 - 8)) \right) \right) = 1 - \left(-1 + \left(4 - (-1 + (-1)) \right) \right) = 1 - \left(-1 + (4 - (-1 - 1)) \right) = 1 - \left(-1 + (4 - (-2)) \right) = 1 - (-1 + 6) = 1 - 5 = 1 + 3 = \mathbf{-4}$$

2.5 Aufgaben zu Kapitel 2.9 bis 2.10

- Multiplikation mit negativen Zahlen

- Division zweier Zahlen

I. Berechne:

a) $3 \cdot (-5) = \mathbf{-15}$

b) $(-4) \cdot 8 = \mathbf{-32}$

c) $(-4) \cdot (-4) = \mathbf{16}$

d) $-(-2) \cdot (-3) = 2 \cdot (-3) = \mathbf{-6}$

e) $(-5) \cdot (-6) = \mathbf{30}$

f) $8 \cdot (-4) + (-2) \cdot (-3) = -32 + 6 = \mathbf{-26}$

g) $(-3) \cdot 5 + (-7) \cdot (-4) = -15 + 28 = \mathbf{13}$

h) $(-3) \cdot (-2) + 2 \cdot (-1) + 5 = 6 + (-2) + 5 = 4 + 5 = \mathbf{9}$

i) $(-5) \cdot 3 + 9 + (-4) \cdot (-2) = -15 + 9 + 8 = -6 + 8 = \mathbf{2}$

j) $(-3) \cdot 8 - (-7) \cdot (-5) = -24 - 35 = \mathbf{-59}$

k) $(-1) \cdot (-1) = \mathbf{1}$

l) $(-1) \cdot 1 = \mathbf{-1}$

m) $1 \cdot (-1) = \mathbf{-1}$

n) $(-1) \cdot 8 + (-1) \cdot 5 = -8 - 5 = \mathbf{-13}$

o) $(-1) \cdot 9 + (-1) \cdot 4 + 7 + (-1) \cdot 3 = -9 - 4 + 7 - 3 = -13 + 7 - 3 = -6 - 3 = \mathbf{-9}$

II. Berechne:

1. $54 : (-3) = -18$

2. $(-48) : 8 = -6$

3. $(-56) : (-7) = 8$

4. $(-144) : 6 = -24$

5. $(-54) : (-6) = 9$

6. $\frac{-3}{4} = -\frac{3}{4}$

7. $\frac{10}{-16} = -\frac{5}{8}$

8. $\frac{-10}{-12} = \frac{5}{6}$

9. $\frac{-2+3-7+4}{5-8+2-6+1} = \frac{1-3}{-3-4+1} = \frac{-2}{-6} = \frac{1}{3}$

10. $\frac{3-8+1-11+5}{5-8+2-6+1-(-10)} = \frac{-5-10+5}{-3-4+11} = \frac{-15+5}{-7+11} = \frac{-10}{4} = -\frac{5}{2}$

11. $\frac{1+4-1+4-2+1-3+5}{1-7+3-20+50-8} = \frac{5+3-1+2}{-6-17+42} = \frac{8-3}{-23+42} = \frac{5}{19}$

12. $\frac{-3-1+4+1-5+9-2-7}{2+2-3-6-0+6-8} = \frac{-4+5+4-9}{4-9-2} = \frac{1-5}{-5-2} = \frac{-4}{-7} = \frac{4}{7}$

13. $\frac{-4-7-7+1-2+1-3}{6-9-3+1-4+7-2} = \frac{-11-6-1-2}{-3-2+3} = \frac{-17-3}{-5+3} = \frac{-21}{-2} = \frac{21}{2}$

14. $\frac{3-8+1-11+5}{5-8+2-6+1-(-10)} = \frac{-5-10+5}{-3-4+11} = \frac{-15+5}{-7+11} = \frac{-10}{4} = -\frac{5}{2}$

15. $\frac{2-7+1-8+2-8+1-8}{3-1-6-2-2+7+7-7} = \frac{-5-7-6-7}{2-8+5} = \frac{-12-13}{-6+5} = \frac{-25}{-1} = 25$

16. $\frac{-3}{4} + \frac{2}{-5} = -\frac{3}{4} - \frac{2}{5} = -\frac{15}{20} - \frac{8}{20} = -\frac{23}{20}$

17. $\frac{-4}{-5} + \frac{2}{-8} = \frac{4}{5} - \frac{1}{4} = \frac{16}{20} - \frac{5}{20} = \frac{11}{20}$

18. $\frac{4}{-7} - \frac{-3}{5} = -\frac{4}{7} + \frac{3}{5} = -\frac{20}{35} + \frac{21}{35} = \frac{1}{35}$

19. $\frac{-1}{9} - \frac{1}{-11} = -\frac{1}{9} + \frac{1}{11} = -\frac{11}{99} + \frac{9}{99} = -\frac{2}{99}$

20. $\frac{5}{-13} + \frac{4}{-5} = -\frac{5}{13} - \frac{4}{5} = -\frac{25}{65} - \frac{52}{65} = -\frac{77}{65}$

2.6 Aufgaben zu Kapitel 2.11

- Potenzen

I. Berechne:

$$\text{a) } 2^{-5} = \frac{1}{2^5} = \frac{1}{2^3} \cdot \frac{1}{2^2} = \frac{1}{8} \cdot \frac{1}{4} = \frac{1}{32}$$

$$\text{b) } (-3)^{-4} = -\frac{1}{3^4} = -\frac{1}{3^2} \cdot \frac{1}{3^2} = -\frac{1}{9} \cdot \frac{1}{9} = -\frac{1}{81}$$

$$\text{c) } (-3)^{-3} = -\frac{1}{3^3} = -\frac{1}{3^2} \cdot \frac{1}{3^1} = -\frac{1}{9} \cdot \frac{1}{3} = -\frac{1}{27}$$

$$\text{d) } 5^{-4} = \frac{1}{5^4} = \frac{1}{5^2} \cdot \frac{1}{5^2} = \frac{1}{25} \cdot \frac{1}{25} = \frac{1}{125} \cdot \frac{1}{5} = \frac{1}{625}$$

$$\text{e) } 2^{-10} = \frac{1}{2^{10}} = \frac{1}{2^5} \cdot \frac{1}{2^5} = \frac{1}{32} \cdot \frac{1}{32} = \frac{1}{64} \cdot \frac{1}{16} = \frac{1}{128} \cdot \frac{1}{8} = \frac{1}{256} \cdot \frac{1}{4} = \frac{1}{512} \cdot \frac{1}{2} = \frac{1}{1024}$$

$$\text{f) } 10^{-3} = \frac{1}{10^3} = \frac{1}{1000}$$

$$\text{g) } 2^9 \cdot 2^8 \cdot 2^{-14} = 2^{9+8-14} = 2^{17-14} = 2^3 = 8$$

$$\text{h) } 2^7 \cdot 3^7 \cdot 2^{-5} \cdot 3^{-5} = 2^{7-5} \cdot 3^{7-5} = 2^2 \cdot 3^2 = 4 \cdot 9 = 36$$

$$\text{i) } 2^{93} \cdot 2^{27} \cdot 2^{-42} \cdot 2^{-69} = 2^{93+27-42-69} = 2^{120-111} = 2^9 = 2^5 \cdot 2^4 = 32 \cdot 16 = 64 \cdot 8 = 128 \cdot 4 = 256 \cdot 2 = 512$$

$$\text{j) } 2^{23} \cdot 3^2 \cdot 2^{-3} \cdot 2^{-16} = 2^{23-3-16} \cdot 3^2 = 2^4 \cdot 3^2 = 16 \cdot 9 = 144$$

$$\text{k) } 2^5 + 2^5 \cdot 3^5 \cdot 3^{-5} = 32 + 32 \cdot 3^{5-5} = 32 + 32 \cdot 3^0 = 32 + 32 \cdot 1 = 64$$

$$\text{l) } 2^{-3} \cdot 3^2 = \frac{3^2}{2^3} = \frac{9}{8}$$

II. Berechne:

$$\text{a) } \frac{3^5 \cdot 3^8}{3^2 \cdot 3^4} = \frac{3^{5+8}}{3^{2+4}} = \frac{3^{13}}{3^6} = 3^{13-6} = 3^7 = 3^4 \cdot 3^3 = 81 \cdot 27 = 243 \cdot 9 = 2.187$$

$$\text{b) } \frac{2^7 \cdot 2^4}{2^5} = \frac{2^{7+4}}{2^5} = \frac{2^{11}}{2^5} = 2^{11-5} = 2^6 = 2^3 \cdot 2^3 = 8 \cdot 8 = 64$$

$$\text{c) } \frac{2^5 \cdot 3^5}{2^3 \cdot 3^2} = 2^{5-3} \cdot 3^{5-2} = 2^2 \cdot 3^3 = 4 \cdot 27 = 108$$

$$\text{d) } \frac{(2^3)^2}{2^2 \cdot 2^3} = \frac{2^{3 \cdot 2}}{2^{2+3}} = \frac{2^6}{2^5} = 2^{6-5} = 2^1 = 2$$

$$\text{e) } \frac{(2^2)^2 \cdot (3^2)^2}{2^2 \cdot 3^2} = 2^{2 \cdot 2-2} \cdot 3^{2 \cdot 2-2} = 2^{4-2} \cdot 3^{4-2} = 2^2 \cdot 3^2 = 4 \cdot 9 = 36$$

$$\text{f) } \frac{2^4 \cdot 3^4 \cdot 2^5}{3^3 \cdot (3^2)^2} = 2^{4+5} \cdot 3^{4-3-2 \cdot 2} = 2^9 \cdot 3^{1-4} = 2^9 \cdot 3^{-3} = \frac{2^9}{3^3} = \frac{512}{27}$$

2.7 Aufgaben zu Kapitel 2.12

- Wissenschaftliche Notation von Zahlen

I. Berechne:

- a) $3,517 \cdot 10^3 = \mathbf{3.517}$
- b) $1,4142135 \cdot 10^{-3} = \mathbf{0,001.414.213.5}$
- c) $1,73205 \cdot 10^{-1} = \mathbf{0,173.205}$
- d) $3,14159 \cdot 10^{12} = \mathbf{3.141.590.000.000}$
- e) $2,7182818 \cdot 10^{-7} = \mathbf{0,000.000.271.828.18}$
- f) $3,1622776 \cdot 10^{-1} = \mathbf{0,316.227.76}$
- g) $8660,254 \cdot 10^{-4} = \mathbf{0,866.025.4}$
- h) $7071067,8 \cdot 10^{-8} = \mathbf{0,070.710.678}$
- i) $0,30102999 \cdot 10^{-6} = \mathbf{0,000.000.301.029.99}$
- j) $693,147180 \cdot 10^7 = \mathbf{6.931.471.800}$

II. Schreibe in wissenschaftlicher Notation:

- a) $7.325 = \mathbf{7,325 \cdot 10^3}$
- b) $1.358.732 = \mathbf{1,358.732 \cdot 10^6}$
- c) $972.144 = \mathbf{9,721.44 \cdot 10^5}$
- d) $0,141.537 = \mathbf{1,415.37 \cdot 10^{-1}}$
- e) $0,053.428.9 = \mathbf{5,342.89 \cdot 10^{-2}}$
- f) $0,000.024.698 = \mathbf{2,469.8 \cdot 10^{-5}}$
- g) $14.031.879 = \mathbf{1,403.187.9 \cdot 10^7}$
- h) $0,001.710.927 = \mathbf{1,710.927 \cdot 10^{-3}}$
- i) $0,000.000.000.08 = \mathbf{8,0 \cdot 10^{-11}}$
- j) $432.177.605 = \mathbf{4,321.776.05 \cdot 10^8}$

3 Buchstabenrechnung

3.1 Aufgaben zu Kapitel 3.1

- Addition und Subtraktion von Buchstaben

I. Berechne:

- a) $8a - 5a = \mathbf{3a}$
- b) $-5a + 8a = \mathbf{3a}$

- c) $-7a + 3a = -4\mathbf{a}$
- d) $13a - 19a = -6\mathbf{a}$
- e) $-21a + 8a = -13\mathbf{a}$
- f) $-8b - 5b = -13\mathbf{b}$
- g) $8b - 5b + 7b - 20b = 3b - 13b = -10\mathbf{b}$
- h) $-9c - 4c + 11c - 17c = -13c - 6c = -19\mathbf{c}$
- i) $-(9a + 4a) - (-11a + 17a) = -13a - 6a = -19\mathbf{a}$
- j) $-(9a - 4a + 11a - 17a) = -(5a - 6a) = -(-a) = \mathbf{a}$
- k) $4x - 9x = -5\mathbf{x}$
- l) $-5x + 18x = 13\mathbf{x}$
- m) $12x - 7x = 5\mathbf{x}$
- n) $8x - 5y - 3x + 2y = 5\mathbf{x} - 3\mathbf{y}$
- o) $x - 6y + 4y - 3x = -2\mathbf{x} - 2\mathbf{y}$
- p) $8a - 5b + 2c - 3a + 7b - 4c = 5\mathbf{a} + 2\mathbf{b} - 2\mathbf{c}$

II. Berechne:

- a) $(4a + 5b) + (2a + 3b) = 4a + 5b + 2a + 3b = 6\mathbf{a} + 8\mathbf{b}$
- b) $(4a + 5b) + (2a - 3b) = 4a + 5b + 2a - 3b = 6\mathbf{a} + 2\mathbf{b}$
- c) $(4a + 5b) - (2a - 3b) = 4a + 5b - 2a + 3b = 2\mathbf{a} + 8\mathbf{b}$
- d) $(4a + 5b) - (-2a + 3b) = 4a + 5b + 2a - 3b = 6\mathbf{a} + 2\mathbf{b}$
- e) $(4a + 5b) - (-2a - 3b) = 4a + 5b + 2a + 3b = 6\mathbf{a} + 8\mathbf{b}$
- f) $-(9a - 4b) - (-11a + 17b) = -9a + 4b + 11a - 17b = 2\mathbf{a} - 13\mathbf{b}$
- g) $7a - (4b + 6a) = 7a - 4b - 6a = \mathbf{a} - 4\mathbf{b}$
- h) $-(2a - (6a - 8b)) = -(2a - 6a + 8b) = -2a + 6a - 8b = 4\mathbf{a} - 8\mathbf{b}$
- i) $5a - \left((-3b + 2a) - (7a + (-9b)) \right) = 5a - (-3b + 2a - 7a - (-9b)) = 5a - (6b - 5a) = 5a - 6b + 5a = 10\mathbf{a} - 6\mathbf{b}$
- j) $x - \left[2y - 3x + \left(4x - (5y - 6x + (7x - 8y)) \right) \right] = x - 2y + 3x - (4x - 5y + 6x - (7x - 8y)) = 4x - 2y - (10x - 5y - 7x + 8y) = 4x - 2y - (3x + 3y) = 4x - 2y - 3x - 3y = \mathbf{x} - 5\mathbf{y}$

III. Berechne:

- a) $6c - 3c = \mathbf{3c}$
- b) $7d - 11d = -\mathbf{4d}$
- c) $-5x + 8x = \mathbf{3x}$
- d) $-4y - 7y = -\mathbf{11y}$
- e) $5c - 2b + 4a + 1 - 3b + 2a - 7c - 3 = \mathbf{6a - 5b - 2c - 2}$
- f) $-8y + 2z - x - 4z + 2y + 5x = \mathbf{4x - 6y - 2z}$
- g) $4w - 2v + 8u - 5w - 11u = -\mathbf{3u - 2v - w}$
- h) $3n - 2m + 2n - 4n - 9n = (3 + 2 - 4 - 9)n - 2m = (5 - 13)n - 2m = -\mathbf{2m - 8n}$

IV. Berechne:

- a) $2a - 3b + (4b - a) - (7a - 2b - 3b) = 2a - 3b + 4b - a - (7a - 5b) = 2a + b - a - 7a + 5b = (2 - 1 - 7)a + (-1 + 5)b = -\mathbf{6a + 4b}$
- b) $2b - 3c + 4a - (-6c + 2a - 2c + 3b - 7a) = 2b - 3c + 4a - (-8c - 5a + 3b) = 2b - 3c + 4a + 8c + 5a - 3b = \mathbf{9a - 1b + 5c}$
- c) $-(-2x + 7y - 3x) + (5y - 6x + 2x - 3y) - 3x + 9y = -(-5x + 7y) + (2y - 4x) - 3x + 9y = 5x - 7y + 2y - 4x - 3x + 9y = (5 - 4 - 3)x + (-7 + 2 + 9)y = -\mathbf{2x + 4y}$
- d) $-3y - 4x - (2y - x + 3y) = -3y - 4x - (5y - x) = -3y - 4x - 5y + x = (-4 + 1)x + (-3 - 5)y = -\mathbf{3x - 8y}$
- e) $6 - 3b + 2a - (2b - 3a - (4a - 3b)) = 6 - 3b + 2a - (2b - 3a - 4a + 3b) = 6 - 3b + 2a - (5b - 7a) = 6 - 3b + 2a - 5b + 7a = (2 + 7)a + (-3 - 5)b + 6 = \mathbf{9a - 8b + 6}$
- f) $-7x + 3 - (3x + 1 - 7x + 3) = -7x + 3 - (-4x + 4) = -7x + 3 + 4x - 4 = -\mathbf{3x - 1}$
- g) $2 - (3v - 5w) + (2w - 4 - v + 5w + 1) = 2 - 3v + 5w + (7w - 3 - v) = 2 - 3v + 5w + (7w - 3 - v) = 2 - 3v + 5w + 7w - 3 - v = (-3 - 1)v + (5 + 7)w - 1 = -\mathbf{4v + 12w - 1}$
- h) $(2 - (u - 3v) + 5u) - (3v + 1 - (-2v + 3u)) = (2 - u + 3v + 5u) - (3v + 1 + 2v - 3u) = (2 + 4u + 3v) - (5v + 1 - 3u) = 2 + 4u + 3v - 5v - 1 + 3u = (4 + 3)u + (3 - 5)v + 1 = \mathbf{7u - 2v + 1}$

3.2 Aufgaben zu Kapitel 3.2 bis 3.4

- Reine Multiplikation von Buchstaben
- Multiplikation von Summen und Differenzen
- Gemischte Beispiele

I. Multipliziere aus:

- a) $x \cdot (x + 5) = \mathbf{x^2 + 5x}$
- b) $(x + 3) \cdot x = \mathbf{x^2 + 3x}$
- c) $x \cdot (x - 7) = \mathbf{x^2 - 7x}$

- d) $a \cdot (8 - a) = -\mathbf{a^2} + \mathbf{8a}$
- e) $-3 \cdot (x - 12) = -\mathbf{3x} + \mathbf{36}$
- f) $a \cdot (-x - 5) = -\mathbf{ax} - \mathbf{5a}$
- g) $-a \cdot (-x + 5) = \mathbf{ax} - \mathbf{5a}$
- h) $x \cdot (8x - 5z - 3 + 2y) = \mathbf{8x^2} + \mathbf{2xy} - \mathbf{5xz} - \mathbf{3x}$
- i) $(a - 2b + 3c) \cdot a = \mathbf{a^2} - \mathbf{2ab} + \mathbf{3ac}$
- j) $(a - b + c) \cdot (-a) = -\mathbf{a^2} + \mathbf{ab} - \mathbf{ac}$
- k) $(a + 3) \cdot (b + 4) = \mathbf{ab} + \mathbf{4a} + \mathbf{3b} + \mathbf{12}$
- l) $(a + 2) \cdot (-b + 5) = -\mathbf{ab} + \mathbf{5a} - \mathbf{2b} + \mathbf{10}$
- m) $(x + 6) \cdot (x + 3) = x^2 + 3x + 6x + 18 = \mathbf{x^2} + \mathbf{9x} + \mathbf{18}$
- n) $(a + 1) \cdot (b + 2) = \mathbf{ab} + \mathbf{2a} + \mathbf{b} + \mathbf{2}$
- o) $(2x + 3) \cdot (5x - 7) = 10x^2 - 14x + 15x - 21 = \mathbf{10x^2} + \mathbf{x} - \mathbf{21}$
- p) $(2x + 3) \cdot (3x - 5) = 6x^2 - 10x + 9x - 15 = \mathbf{6x^2} - \mathbf{x} - \mathbf{15}$

II. Multipliziere aus:

- a) $(x + 1) \cdot (x + 2) = x^2 + 2x + x + 2 = \mathbf{x^2} + \mathbf{3x} + \mathbf{2}$
- b) $(x + 4) \cdot (x + 5) = x^2 + 5x + 4x + 20 = \mathbf{x^2} + \mathbf{9x} + \mathbf{20}$
- c) $(x + 2) \cdot (x - 3) = x^2 - 3x + 2x - 6 = \mathbf{x^2} - \mathbf{x} - \mathbf{6}$
- d) $(x - 3) \cdot (x + 4) = x^2 + 4x - 3x - 12 = \mathbf{x^2} + \mathbf{x} - \mathbf{12}$
- e) $(x - 5) \cdot (x - 1) = x^2 - x - 5x + 5 = \mathbf{x^2} - \mathbf{6x} + \mathbf{5}$
- f) $(-x + 3) \cdot (x - 4) = -x^2 + 4x + 3x - 12 = -\mathbf{x^2} + \mathbf{7x} - \mathbf{12}$
- g) $(-x - 1) \cdot (-x + 2) = x^2 - 2x + x - 2 = \mathbf{x^2} - \mathbf{x} - \mathbf{2}$
- h) $(-x - 3) \cdot (-x - 7) = x^2 + 7x + 3x + 21 = \mathbf{x^2} + \mathbf{10x} + \mathbf{21}$
- i) $(a - 2) \cdot (b + 5) = \mathbf{ab} + \mathbf{5a} - \mathbf{2b} - \mathbf{10}$
- j) $(u + 1) \cdot (-v + 6) = -\mathbf{uv} + \mathbf{6u} - \mathbf{v} + \mathbf{6}$
- k) $(x - 4) \cdot (b - 3) = \mathbf{bx} - \mathbf{4b} - \mathbf{3x} + \mathbf{12}$
- l) $(z + 2) \cdot (w - 1) = \mathbf{wz} + \mathbf{2w} - \mathbf{z} - \mathbf{2}$
- m) $(x - 7) \cdot (y + 6) = \mathbf{xy} + \mathbf{6x} - \mathbf{7y} - \mathbf{42}$
- n) $(k - 2) \cdot (k - 3) = k^2 - 3k - 2k + 6 = \mathbf{k^2} - \mathbf{5k} + \mathbf{6}$
- o) $(m + 4) \cdot (n - 4) = \mathbf{mn} - \mathbf{4m} + \mathbf{4n} - \mathbf{16}$
- p) $(a + 3) \cdot (a + 3) = \mathbf{a^2} + \mathbf{6a} + \mathbf{9}$

III. Multipliziere aus:

- a) $(3x + 2) \cdot (4x + 5) = 12x^2 + 15x + 8x + 10 = \mathbf{12x^2 + 23x + 10}$
- b) $(4x + 3) \cdot (2x + 1) = 8x^2 + 4x + 6x + 3 = \mathbf{8x^2 + 10x + 3}$
- c) $(5x + 2) \cdot (3x - 4) = 15x^2 - 20x + 6x - 8 = \mathbf{15x^2 - 14x - 8}$
- d) $(2x - 3) \cdot (y + 4) = \mathbf{2xy + 4x - 3y - 12}$
- e) $(3y - 5) \cdot (2x - 6) = \mathbf{6xy - 10x - 18y + 30}$
- f) $(-2x + 3y) \cdot (5x - 4y) = -10x^2 + 8xy + 15xy - 12y^2 = \mathbf{-10x^2 + 23xy - 12^2}$
- g) $(-x - y) \cdot (-2x + y) = 2x^2 - xy + 2xy - y^2 = \mathbf{2x^2 + xy - y^2}$
- h) $(-a - 2b) \cdot (-a - 6b) = a^2 + 6ab + 2ab + 12b^2 = \mathbf{a^2 + 8ab + 12b^2}$
- i) $(3a - 2b) \cdot (5a + b) = 15a^2 + 3ab - 10ab - 2b^2 = \mathbf{15a^2 - 7ab - 2b^2}$
- j) $(u + v) \cdot (-u + 4v) = -u^2 + 4uv - uv + 4v^2 = \mathbf{-u^2 + 3uv + 4v^2}$
- k) $(2u - 4w) \cdot (3v - 5w) = \mathbf{6uv - 10uw - 12vw + 20w^2}$
- l) $(w + 2) \cdot (z - 1) = \mathbf{wz - w + 2z - 2}$
- m) $(5x - 6y) \cdot (2x + 3y) = 10x^2 + 15xy - 12xy - 18y^2 = \mathbf{10x^2 + 3xy - 18y^2}$
- n) $(a - 2b + 3c) \cdot (2a - b + 5) = 2a^2 - ab + 5a - 4ab + 2b^2 - 10b + 6ac - 3bc + 15c = \mathbf{2a^2 - 5ab + 6ac + 2b^2 - 3bc + 5a - 10b + 15c}$
- o) $(2m + 3n) \cdot (3m - 2n) = 6m^2 - 4mn + 9mn - 6n^2 = \mathbf{6m^2 + 5mn - 6n^2}$
- p) $(2a + 3) \cdot (2a - 3) = \mathbf{4a^2 - 9}$

IV. Multipliziere aus:

- a) $\left(\frac{3}{2}x + \frac{1}{2}y\right) \cdot \left(\frac{5}{2}x - \frac{1}{2}y\right) = \frac{15}{4}x^2 - \frac{3}{4}xy + \frac{5}{4}xy - \frac{1}{4}y^2 = \frac{15}{4}\mathbf{x^2} + \frac{1}{2}\mathbf{xy} - \frac{1}{4}\mathbf{y^2}$
- b) $\left(-\frac{2}{5}x + \frac{1}{3}y\right) \cdot \left(\frac{3}{4}x - \frac{5}{6}y\right) = -\frac{6}{20}x^2 + \frac{10}{30}xy + \frac{3}{12}xy - \frac{5}{18} = -\frac{3}{10}x^2 + \frac{1}{3}xy + \frac{1}{4}xy - \frac{5}{18} = \mathbf{-\frac{3}{10}x^2 + \frac{7}{12}xy - \frac{5}{18}y^2}$
- c) $x \cdot \left(1 + \frac{1}{2}x \cdot \left(1 + \frac{1}{3}x\right)\right) = x \cdot \left(1 + \frac{1}{2}x + \frac{1}{6}x^2\right) = \frac{1}{6}\mathbf{x^3} + \frac{1}{2}\mathbf{x^2} + \mathbf{x}$
- d) $(0, 2x - 1, 3y) \cdot (0, 5x + 4y) = 0, 1x^2 + 0, 8xy - 0, 65xy - 5, 2y^2 = \mathbf{0, 1x^2 - 0, 15xy - 5, 2y^2}$
- e) $(x - 1) \cdot (1 + x + x^2 + x^3) = x + x^2 + x^3 + x^4 - 1 - x - x^2 - x^3 = \mathbf{x^4 - 1}$
- f) $(x + 1) \cdot (1 - x + x^2 - x^3) = x - x^2 + x^3 - x^4 + 1 - x + x^2 - x^3 = \mathbf{-x^4 + 1}$
- g) $(4x^2 - 9y^2) \cdot (4x^2 + 9y^2) = (4x^2)^2 - (9y^2)^2 = \mathbf{16x^4 - 81y^4}$
- h) $(u^2 - 3v) \cdot (-2u - 5v^2) = \mathbf{-5u^2v^2 - 2u^3 + 15v^3 + 6uv}$

- i) $(2a - 3c) \cdot (4b + 2c) = 8ab + 4ac - 12bc - 6c^2$
- j) $(5u - 3v + 2w) \cdot (-u + 2v + 3w) = -5u^2 + 10uv + 15uw + 3uv - 6v^2 - 9vw - 2uw + 4vw + 6w^2 = -5u^2 + 13uv + 13uw - 6v^2 - 5vw + 6w^2$
- k) $(4x - 2y + 3) \cdot (5x - 3y + 2) = 20x^2 - 12xy + 8x - 10xy + 6y^2 - 4y + 15x - 9y + 6 = 20x^2 - 22xy + 6y^2 + 23x - 13y + 6$
- l) $(x^2 - 3x + 2) \cdot (3y - 4) = 3x^2y - 4x^2 - 9xy + 12x + 6y - 8$
- m) $(3x - 6y + 2z) \cdot (5x + 3y) = 15x^2 + 9xy - 30xy - 18y^2 + 10xz + 6yz = 15x^2 - 21xy + 10xz - 18y^2 + 6yz$
- n) $(a - 2b + 3c) \cdot (a - 2b + 3c) = a^2 - 4ab + 6ac + 4b^2 - 12bc + 9c^2 = a^2 - 4ab + 6ac + 4b^2 - 12bc + 9c^2$
- o) $(3u + 2v + 1) \cdot (3u + 2v - 1) = 9u^2 + 6uv - 3u + 6uv + 4v^2 - 2v + 3u + 2v - 1 = 9u^2 + 12uv + 4v^2 - 1$
- p) $(4a - 3) \cdot (4a - 3) = (4a)^2 - 2 \cdot 3 \cdot 4a + 3^2 = 16a^2 - 24a + 9$

V. Multipliziere aus:

- a) $3x \cdot (2x + 5) \cdot x = 3x^2 \cdot (2x + 5) = 6x^3 + 15x^2$
- b) $(3x + 5) \cdot (2x + 3) \cdot x = (6x^2 + 9x + 10x + 15) \cdot x = (6x^2 + 19x + 15) \cdot x = 6x^3 + 19x^2 + 15x$
- c) $(x + 5) \cdot (x + 3) \cdot (x - 7) = (x^2 + 8x + 15) \cdot (x - 7) = x^3 - 7x^2 + 8x^2 - 56x + 15x - 105 = x^3 + x^2 - 41x - 105$
- d) $(x + y + 5) \cdot (x + 2y + 3) = x^2 + 2xy + 3x + xy + 2y^2 + 3y + 5x + 10y + 15 = x^2 + 3xy + 2y^2 + 8x + 13y + 15$
- e) $(2x - 3y + 5) \cdot (5x + y - 1) = 10x^2 + 2xy - 2x - 15xy - 3y^2 + 3y + 25x + 5y - 5 = 10x^2 - 13xy - 3y^2 + 23x + 8y - 5$
- f) $(x - 1) \cdot (1 + x + x^2) = x + x^2 + x^3 - 1 - x - x^2 = x^3 - 1$
- g) $(x - 1) \cdot (x + 2) \cdot (x - 3) = (x^2 + 2x - x - 2) \cdot (x - 3) = (x^2 + x - 2) \cdot (x - 3) = x^3 - 3x^2 + x^2 - 3x - 2x + 6 = x^3 - 2x^2 - 5x + 6$
- h) $(3x + 1) \cdot (5x - 2) \cdot (2x - 3) = (15x^2 - 6x + 5x - 2) \cdot (2x - 3) = (15x^2 - x - 2) \cdot (2x - 3) = 30x^3 - 45x^2 - 2x^2 + 3x - 4x + 6 = 30x^3 - 47x^2 - x + 6$
- i) $(x^2 - x) \cdot (1 + x + x^2) = x^2 + x^3 + x^4 - x - x^2 - x^3 = x^4 - x$
- j) $(x^2 - 1) \cdot (1 + x + x^2 + x^3) = x^2 + x^3 + x^4 + x^5 - 1 - x - x^2 - x^3 = x^5 + x^4 - x - 1$
- k) $(a + 3) \cdot (b + 4) \cdot (c + 5) = (ab + 4a + 3b + 12) \cdot (c + 5) = abc + 5ab + 4ac + 3bc + 20a + 15b + 12c + 60$
- l) $(a + b + 2) \cdot (-b + 3 + c) = -ab + ac - b^2 + bc + 3a + b + 2c + 6$
- m) $(2a - 3b + 5c) \cdot (a + b - 3 + c) = 2a^2 + 2ab - 6a + 2ac - 3ab - 3b^2 + 9b - 3bc + 5ac + 5bc - 15c + 5c^2 = 2a^2 - ab + 7ac - 3b^2 + 2bc + 5c^2 - 6a + 9b - 15c$
- n) $(a + 1) \cdot (b + 2) \cdot (c - 3) = (ab + 2a + b + 2) \cdot (c - 3) = abc - 3ab + 2ac + bc - 6a - 3b + 2c - 6$
- o) $(x + 1) \cdot (x + 1) \cdot (x + 1) \cdot (x + 1) = (x + 1)^2 \cdot (x + 1)^2 = (x^2 + 2x + 1) \cdot (x^2 + 2x + 1) = x^4 + 2x^3 + x^2 + 2x^3 + 4x^2 + 2x + x^2 + 2x + 1 = x^4 + 4x^3 + 6x^2 + 4x + 1$

VI. Multipliziere aus:

- a) $(x + 3)^2 = \mathbf{x^2 + 6x + 9}$
- b) $(x - 1) \cdot (x + 2) + (x + 3) \cdot (x + 1) = x^2 + x - 2 + x^2 + 4x + 3 = \mathbf{2x^2 + 5x + 1}$
- c) $(x + 5)^2 + 3 \cdot (2x + 3) = x^2 + 10x + 25 + 6x + 9 = \mathbf{x^2 + 16x + 34}$
- d) $2 \cdot (x + 5) \cdot (x + 3) - (x - 7) \cdot (2x + 1) = 2 \cdot (x^2 + 8x + 15) - (2x^2 + x - 14x - 7) = 2x^2 + 16x + 30 - (2x^2 - 13x - 7) = 2x^2 + 16x + 30 - 2x^2 + 13x + 7 = \mathbf{29x + 37}$
- e) $(2x + y)^2 - (x - 3y)^2 = 4x^2 + 4xy + y^2 - (x^2 - 6xy + 9y^2) = 4x^2 + 4xy + y^2 - x^2 + 6xy - 9y^2 = \mathbf{3x^2 + 10xy - 8y^2}$
- f) $(2a - 3b) \cdot (3a + b) - 2 \cdot (3a + b) \cdot (a - b) = 6a^2 + 2ab - 9ab - 3b^2 - 2 \cdot (3a^2 - 3ab + ab - b^2) = 6a^2 - 7ab - 3b^2 - 2 \cdot (3a^2 - 2ab - b^2) = 6a^2 - 7ab - 3b^2 - 6a^2 + 4ab + 2b^2 = \mathbf{-3ab - b^2}$

3.3 Aufgaben zu Kapitel 3.5 und 3.6

- Die binomischen Formeln

- Anwendungsbeispiele für die 1. und 2. binomische Formel

I. Benutze die *binomische Formeln*:

- a) $(x + 5)^2 = \mathbf{x^2 + 10x + 25}$
- b) $(x + 2)^2 = \mathbf{x^2 + 4x + 4}$
- c) $(x + 3) \cdot (x + 3) = \mathbf{x^2 + 6x + 9}$
- d) $(x + y)^2 = \mathbf{x^2 + 2xy + y^2}$
- e) $(x + 3y)^2 = \mathbf{x^2 + 6xy + 9y^2}$
- f) $(5x + 7y)^2 = \mathbf{25x^2 + 70xy + 49y^2}$
- g) $(3u + 2v)^2 = \mathbf{9u^2 + 12uv + 4v^2}$
- h) $(a + 5b)^2 = \mathbf{a^2 + 10ab + 25b^2}$
- i) $(x + 5y)^2 = \mathbf{x^2 + 10xy + 25y^2}$
- j) $(x - 5y)^2 = \mathbf{x^2 - 10xy + 25y^2}$
- k) $(2x - y)^2 = \mathbf{4x^2 - 4xy + y^2}$
- l) $(-2x + 3y)^2 = \mathbf{4x^2 - 12xy + 9y^2}$
- m) $(-x - 7)^2 = \mathbf{x^2 + 14x + 49}$
- n) $(2a - 3b)^2 = \mathbf{4a^2 - 12ab + 9b^2}$
- o) $(7x - 3)^2 = \mathbf{49x^2 - 42x + 9}$

II. Berechne:

- a) $(5 \cdot (x + 1))^2 = 25 \cdot (x^2 + 2x + 1) = \mathbf{25x^2 + 50x + 25}$
- b) $(4 \cdot (x - 3))^2 = 16 \cdot (x^2 - 6x + 9) = \mathbf{16x^2 - 96x + 144}$
- c) $(3 \cdot (5x + 2))^2 = 9 \cdot (25x^2 + 20x + 4) = \mathbf{225x^2 + 180x + 36}$
- d) $(2y \cdot (3x + 1))^2 = 4y^2 \cdot (9x^2 + 6x + 1) = \mathbf{36x^2y^2 + 24xy^2 + 4y^2}$

III. Vereinfache mithilfe der *binomischen Formeln*:

- a) $(x + 3)^2 + (x - 3)^2 = x^2 + 6x + 9 + x^2 - 6x + 9 = \mathbf{2x^2 + 18}$
- b) $(x + 5)^2 + (x - 5)^2 = x^2 + 10x + 25 + x^2 - 10x + 25 = \mathbf{2x^2 + 50}$
- c) $(2x + 1)^2 - (3x - 2)^2 + 3 \cdot (x + 5)^2 = 4x^2 + 4x + 1 - (9x^2 - 12x + 4) + 3 \cdot (x^2 + 10x + 25) = 4x^2 + 4x + 1 - 9x^2 + 12x - 4 + 3x^2 + 30x + 75 = \mathbf{-2x^2 + 46x + 72}$
- d) $(x + 7)^2 + (x - 7)^2 + 2 \cdot (x - 7) \cdot (x + 7) = x^2 + 14x + 49 + x^2 - 14x + 49 + 2 \cdot (x^2 - 49) = 2x^2 + 98 + 2x^2 - 98 = \mathbf{4x^2}$
- e) $(x^2 + 3)^2 - (x^2 - 2)^2 + 4 \cdot (x + 3)^2 = x^4 + 6x^2 + 9 - x^4 + 4x^2 - 4 + 4 \cdot (x^2 + 6x + 9) = 10x^2 + 5 + 4x^2 + 24x + 36 = \mathbf{14x^2 + 24x + 41}$

IV. Berechne mithilfe der *binomische Formeln*:

- a) $101^2 = (100 + 1)^2 = 100^2 + 2 \cdot 100 \cdot 1 + 1^2 = 10.000 + 200 + 1 = \mathbf{10.201}$
- b) $103^2 = (100 + 3)^2 = 100^2 + 2 \cdot 100 \cdot 3 + 3^2 = 10.000 + 600 + 9 = \mathbf{10.609}$
- c) $1.008^2 = (1.000 + 8)^2 = 1.000^2 + 2 \cdot 1.000 \cdot 8 + 8^2 = 1.000.000 + 16.000 + 64 = \mathbf{1.016.064}$
- d) $205^2 = (200 + 5)^2 = 200^2 + 2 \cdot 200 \cdot 5 + 5^2 = 40.000 + 2.000 + 25 = \mathbf{42.025}$
- e) $99^2 = (100 - 1)^2 = 100^2 - 2 \cdot 100 \cdot 1 + 1^2 = 10.000 - 200 + 1 = \mathbf{9.801}$
- f) $94^2 = (100 - 6)^2 = 100^2 - 2 \cdot 100 \cdot 6 + 6^2 = 10.000 - 1.200 + 36 = \mathbf{8.836}$
- g) $993^2 = (1000 - 7)^2 = 1000^2 - 2 \cdot 1000 \cdot 7 + 7^2 = 1.000.000 - 14.000 + 49 = \mathbf{986.049}$
- h) $195^2 = (200 - 5)^2 = 200^2 - 2 \cdot 200 \cdot 5 + 5^2 = 40.000 - 2.000 + 25 = \mathbf{38.025}$

3.4 Aufgaben zu Kapitel 3.7

- Anwendungsbeispiele für 3. binomische Formel

I. Benutze die *dritte binomische Formel*:

- a) $(x + 5) \cdot (x - 5) = \mathbf{x^2 - 25}$
- b) $(x + 2) \cdot (x - 2) = \mathbf{x^2 - 4}$
- c) $(x + 3) \cdot (x - 3) = \mathbf{x^2 - 9}$
- d) $(x + y) \cdot (x - y) = \mathbf{x^2 - y^2}$
- e) $(x + 3y) \cdot (x - 3y) = \mathbf{x^2 - 9y^2}$

- f) $(5x + 7y) \cdot (5x - 7y) = \mathbf{25x^2 - 49y^2}$
 g) $(3u + 2v) \cdot (3u - 2v) = \mathbf{9u^2 - 4v^2}$
 h) $(a + 5b) \cdot (a - 5b) = \mathbf{a^2 - 25b^2}$
 i) $(x + 5y) \cdot (x - 5y) = \mathbf{x^2 - 25y^2}$
 j) $(2x - 3y) \cdot (2x + 3y) = \mathbf{4x^2 - 9y^2}$
 k) $(x + 2y) \cdot (x - 2y) = \mathbf{x^2 - 4y^2}$
 l) $(-2x + 3y) \cdot (-2x - 3y) = \mathbf{4x^2 - 9y^2}$
 m) $(-x + 7) \cdot (x + 7) = (7 - x) \cdot (7 + x) = \mathbf{49 - x^2}$
 n) $(2a + 3b) \cdot (-2a + 3b) = (3b + 2a) \cdot (3b - 2a) = \mathbf{9b^2 - 4a^2}$
 o) $(-7x + 3) \cdot (-7x - 3) = \mathbf{49x^2 - 9}$

II. Berechne:

- a) $5 \cdot (x + 1) \cdot (x - 1) = 5 \cdot (x^2 - 1) = \mathbf{5x^2 - 5}$
 b) $4 \cdot (x - 3) \cdot (x + 3) = 4 \cdot (x^2 - 9) = \mathbf{4x^2 - 36}$
 c) $3 \cdot (-5x + 2) \cdot (5x + 2) = 3 \cdot (4 - 25x^2) = \mathbf{12 - 75x^2}$
 d) $2y \cdot (3x + 1) \cdot (3x - 1) = 2y \cdot (9x^2 - 1) = \mathbf{18x^2y - 2y}$

III. Vereinfache mithilfe der *binomischen Formeln*:

- a) $(x + 3) \cdot (x - 3) + (x - 2) \cdot (x + 2) - (x + 1) \cdot (x - 1) = x^2 - 9 + x^2 - 4 - x^2 + 1 = \mathbf{x^2 - 12}$
 b) $(x + 2)^2 + (x - 2)^2 - (x + 3) \cdot (x - 3) = x^2 + 4x + 4 + x^2 - 4x + 4 - x^2 + 9 = \mathbf{x^2 + 17}$
 c) $(x + 3)^2 + (x - 3)^2 + (x + 3) \cdot (x - 3) = x^2 + 6x + 9 + x^2 - 6x + 9 + x^2 - 9 = \mathbf{3x^2 + 9}$
 d) $\frac{a^2 - b^2}{a + b} = \frac{(a + b) \cdot (a - b)}{(a + b)} = \mathbf{a - b}$
 e) $\frac{a^2 - b^2}{a - b} = \frac{(a + b) \cdot (a - b)}{(a - b)} = \mathbf{a + b}$
 f) $\frac{(x - y)^2}{x^2 - y^2} = \frac{(x - y) \cdot (x - y)}{(x + y) \cdot (x - y)} = \frac{\mathbf{x - y}}{\mathbf{x + y}}$

IV. Berechne mithilfe der *dritten binomischen Formel*:

- a) $107 \cdot 93 = (100 + 7) \cdot (100 - 7) = 100^2 - 7^2 = 10.000 - 49 = \mathbf{9.951}$
 b) $1.006 \cdot 994 = (1.000 + 6) \cdot (1.000 - 6) = 1.000^2 - 6^2 = 1.000.000 - 36 = \mathbf{999.964}$
 c) $101 \cdot 99 = (100 + 1) \cdot (100 - 1) = 100^2 - 1^2 = 10.000 - 1 = \mathbf{9.999}$
 d) $28 \cdot 32 = (30 - 2) \cdot (30 + 2) = 30^2 - 2^2 = 900 - 4 = \mathbf{896}$